

# CyberLab Reference Guide

Field quick-reference for the kit  
capability → tool → command → every option explained

Generated from the repository. Markdown is the source of truth.  
[github.com/project-cyberlab/cyberlab-reference-guide](https://github.com/project-cyberlab/cyberlab-reference-guide)

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fls (*filesystem analysis*)

# CyberLab Reference Guide

A field quick-reference for the kit. You open it mid-mission or mid-exercise when you need a **capability** and do not remember which tool provides it — it routes you to the tool, the exact command, and **every option, explained**.

**Status:** format and scope established; worked example proven end to end. Content build is next. See [docs/PLAN.md](#).

## Start here

| You are...                                                  | Go to                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------------|
| Facing a problem — " <i>I need to carve deleted files</i> " | <b>Capability Index</b> ( <i>being built</i> — see <i>PLAN</i> )  |
| Asking what the kit even has                                | <a href="#">Kit Tool List</a> — 1,006 tools, VM → category → tool |
| Wanting to know how pages are built and why                 | <a href="#">The Format</a>                                        |
| Wanting to see a finished page                              | <code>fls</code> — worked example, all 22 options                 |

## What makes this different

Every DFIR quick-reference in the field — SANS, Jai Minton's, tldr — either organises by tool (so you must already know the tool) or skips flag documentation entirely. The best of them states outright that it "*assumes operator familiarity*."

This guide inverts that:

- **Capability-first.** You search by what you need to *do*.
- **Options completely listed out**, each with *what it does* and *when you'd use it*.
- **Tools attached to commands**, and each tagged with the VM that carries it — so you know whether you can run it on the box in front of you.

## Scope is binding

Only tools in `catalog/kit-tools.json` may be documented. That file is generated from the kit VMs' own upstream manifests:

| Platform         | Tools | Source of truth                                                                  |
|------------------|-------|----------------------------------------------------------------------------------|
| Kali Linux       | 403   | <code>kali-meta</code> <code>debian/control</code> ( <code>kali-tools-*</code> ) |
| REMnux           | 268   | <code>REMnux/docs</code> <code>discover-the-tools</code>                         |
| SIFT Workstation | 162   | <code>teamdfir/sift-saltstack</code>                                             |
| FLARE-VM         | 137   | <code>mandiant/flare-vm</code> <code>config.xml</code>                           |

| Platform       | Tools | Source of truth |
|----------------|-------|-----------------|
| Security Onion | 36    | kit tracker CSV |

Regenerate with `python scripts/build_kit_list.py`.

## Every flag is real

Options are **captured from the running binary** ( `--help` / `man`, inside the kit container), stored raw under `capture/`, and checked by a linter. A flag that cannot be produced from a real binary cannot ship.

This is a direct response to a measured failure in the predecessor project, where fabricated CLI flags reached roughly 44 of 61 modules. Full rationale in [docs/FORMAT.md](#).

## Relationship to cyberlab

Separate repository, deliberately. The [cyberlab training lab](#) is **on hold** and receives no changes from this work.

## Layout

```
catalog/    kit tool list – the binding scope (generated)
reference/  the guide itself, one page per tool, grouped by capability
capture/    raw --help/man output, committed as evidence
docs/       FORMAT.md (how pages are built), PLAN.md (roadmap)
scripts/    build_kit_list.py, and the linter
```

# The Format

How every page in this guide is built, and why. Derived from studying the reference works that already do this well, then fixing what they each get wrong for our use case: **a field guide you open mid-mission when you need a capability and do not remember the tool.**

## 1. What the research said

| Source                                                         | What it does well                                                                                              | What it does not give us                                                               |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <a href="#">tldr-pages style guide</a>                         | Curated examples ( $\leq 8$ ), imperative mood, <code>{{placeholder}}</code> syntax, machine-linted            | Deliberately <b>incomplete</b> — omits most options by design. No capability index.    |
| <a href="#">POSIX Utility Conventions ch.12 + man-pages(7)</a> | Unambiguous synopsis grammar: <code>[]</code> optional, <code>\ </code> exclusive, <code>...</code> repeatable | Reference-shaped, not task-shaped. You must already know the tool's name.              |
| <a href="#">SANS DFIR cheat sheets</a>                         | Organised by <b>investigative task</b> — the right primary axis                                                | Print-first, not greppable, not exhaustive on flags                                    |
| <a href="#">Jai Minton DFIR cheatsheet</a>                     | 200+ scannable anchors; artifact paths and ATT&CK IDs as headings                                              | <i>"Most entries lack detailed flag documentation... assumes operator familiarity"</i> |
| <a href="#">navi / cheat.sh</a>                                | Machine-parseable, taggable, directly executable entries                                                       | No explanation layer at all                                                            |
| <a href="#">Progressive disclosure (Nielsen, 1995)</a>         | Layer information: essentials first, depth on demand                                                           | —                                                                                      |

**The gap we fill:** every existing DFIR quick-reference assumes you already know the tool and its flags. None of them explains the options. That is exactly the layer this guide exists to provide, and it is why "completely listed out" is a hard requirement rather than a stretch goal.

## 2. Two entry points, one dataset

You arrive at this guide in one of two states:

1. **"I have a problem."** — *I need to recover deleted files from this image.* → **Capability Index:** capability → tool → command → options. This is the primary path and the reason the guide exists.
2. **"What do I have?"** — *Is there anything in the kit for Go binaries?* → **Kit Index:** VM → category → tool. This is `catalog/KIT-TOOLS.md`, generated from upstream manifests.

Both views are projections of the same JSON, so they can never disagree.

## 3. Scope is binding

A tool may appear in this guide **only** if it is in `catalog/kit-tools.json`, which is generated from the kit's own VM manifests. If it is not in the kit, it does not get a page — no matter how good the tool is.

Every command entry is tagged with the **VM(s) that carry it**, so in the field you instantly know whether you can actually run it on the box in front of you.

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## 4. The page format

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Every tool page has these sections, in this order. Sections are layered by progressive disclosure: a reader in a hurry stops after §3; a reader who needs precision continues to §4.

### Header

```
# <tool>
**Kit:** REMnux · SIFT    **Category:** Filesystem analysis    **Version:** 4.11.1
**Docs:** <upstream URL>    **Verified:** 2026-07-29 from `cyberlab-aio:v1`
```

The version and the verification date are not decoration — they are the claim that the options below were read off the real binary on a known date.

### 1. Purpose — one line, imperative

List files and directory names in a disk image, including deleted entries.

### 2. Synopsis — POSIX notation, verbatim from the tool

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

### 3. Common invocations — ≤8, task-titled, copy-paste ready

Each example is titled by **the task it accomplishes**, not by the flag it uses, because the reader is searching by intent. Placeholders use tldr syntax so they are visually obvious and machine-checkable:

```
# List deleted files in a partition, recursively, with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}
```

### 4. Options — COMPLETE, one row per flag

Every flag the binary accepts. Not the useful ones — **all** of them.

| Flag            | Argument | What it does                 | When you would use it                         |
|-----------------|----------|------------------------------|-----------------------------------------------|
| <code>-d</code> | —        | Display deleted entries only | Narrowing a recovery hunt to what was removed |

The fourth column is the one that makes this a guide rather than a man page dump. It answers *why would I reach for this*, which is what you actually need under time pressure.

## 5. Gotchas

Real operational traps, each one earned. Examples already known on this project: timeout-guard `vol`, `binwalk`, `clamscan` and `radare2` in any harness — a synthetic memory image once pinned `vol` at 99% CPU indefinitely.

## 6. See also

Sibling tools for the same capability, so a dead end reroutes instead of stalling.

## 5. Verification: why the options can be trusted

The predecessor project shipped **fabricated CLI flags in roughly 44 of its 61 modules** — invented options for `yara`, `clamscan`, PE-bear, `scdbg` and others — because content was generated from a language model rather than read off the tools. That failure is the reason this guide exists in its current form.

So the rule here is mechanical, not aspirational:

1. Run the tool's `--help` / `-h` / man page **inside the kit container or VM**.
2. Store the raw output under `capture/<tool>.help.txt`, committed.
3. Generate/check the options table **against that capture**.
4. `scripts/lint.py` fails the build if a documented flag is absent from the capture, or if a captured flag is missing from the page.

A flag that cannot be produced from a real binary cannot ship. This is the same discipline `tldr-lint` applies to formatting, applied instead to truth.

**Worked proof.** The captured `fls` usage line advertises `[-adDFlhpruvV]`, but the body of the same output documents `-P`, `-B`, `-S` and `-k` as well. Anyone transcribing the synopsis — or working from memory — silently loses four real options. Only a capture catches that.

## 6. Sources and citation validation

Two different claims need two different kinds of evidence, and they must not be confused:

- **What a flag is** → the binary. Captured, committed, linter-enforced (§5). No online source overrides a real `--help`.
- **What a tool is for, and how practitioners actually use it** → authoritative upstream documentation and reputable DFIR publishers.

Acceptable sources for the second kind, in preference order: the project's own docs/repo (sleuthkit.org, volatilityfoundation.org, docs.remnux.org, ericzimmerman.github.io, Microsoft Learn), then recognised practitioner sources (SANS, thedfirreport.com, Red Canary, Mandiant/Unit42), then MITRE ATT&CK for technique mapping.

**Every cited URL is liveness-checked before it ships.** The predecessor project shipped hallucinated citations that looked plausible and 404'd; it already has a hardened, allowlisted, deadline-guarded fetcher (`cyberlab_retrieval.py`, with `verify_url`) built precisely for this, and this guide reuses that discipline. A dead or unreachable citation fails the build rather than degrading quietly.

Version skew is a real hazard: upstream docs describe the *current* release, while the kit runs pinned versions (TSK 4.11.1 in the container vs 4.15.0 in the tracker, for example). Where docs and capture disagree, **the capture wins** and the discrepancy is worth a line in Gotchas.

## 7. Output

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- **Markdown** is the source of truth: greppable, diffable, reviewable, renders on GitHub.
- **PDF/print render** is generated from the markdown for field use. It is a build artifact, never hand-edited, so the two can never drift.



# Implementation Plan

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Goal: a field quick-reference covering the kit's tools, capability-indexed, with every command's options completely listed and explained, verified against real binaries.

Status legend:  done ·  next ·  pending

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## Phase 0 — Foundations

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-  **Scope fixed.** `catalog/kit-tools.json` + `KIT-TOOLS.md` : 1,006 tools, 5 platforms, 54 categories, each from an upstream machine-readable manifest.
-  **Format designed** from six reference works — `docs/FORMAT.md`.
-  **Verification proven.** `cyberlab-aio:v1` on rick has `fls`, `clamscan`, `binwalk`, `yara`, `vol`, `tshark`, `exiftool`, `log2timeline.py`.
-  **Worked example** — `reference/filesystem-analysis/fls.md`, 22 options from a real capture.

## Phase 1 — Capability taxonomy

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The primary index. Draft ~40–60 capabilities phrased as **problems**, not tool names, e.g. *"Recover deleted files from an image"*, *"Find what persisted on a host"*, *"Decode an obfuscated payload"*, *"Prove lateral movement"*.

- Seed from the categories already carried by upstream: REMnux's 11 capability categories, Kali's 12 `kali-tools-*` groups, Security Onion's 9, and SANS investigative-task ordering.
- Map every one of the 1,006 kit tools to  $\geq 1$  capability. Tools that map to none are reported, not silently dropped — an unmapped tool is either a taxonomy gap or genuinely out of scope, and both need a decision.
- **Deliverable:** `reference/INDEX.md` — capability → tools → page links.

**Open question for review:** should the capability taxonomy follow an existing public framework (NIST/DFIR phases, MITRE ATT&CK tactics) for familiarity, or a kit-native one tuned to what we actually carry?

Recommendation: kit-native primary, with ATT&CK IDs as secondary tags where they genuinely apply — forcing forensic tools into attacker tactics distorts both.

## Phase 1b — Mine cyberlab as a source

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The cyberlab training lab already contains a large body of commands and prior exhaustive research. It is **read-only input** here — cyberlab is on hold and receives no changes.

- `scripts/mine_cyberlab.py` : extract every fenced command block from the 61 modules, plus `docs/CURRICULUM_ROADMAP.md`, `docs/TEST_REPORT.md` and `docs/CONTENT_QUALITY_AUDIT.md`, and the `catalog/` research.
- Produce `capture/cyberlab-candidates.json` : tool → observed invocations, with the module each came from.

- These become **candidate invocations** for the "Common invocations" section — real workflows someone already thought through, which is exactly the part that is expensive to invent.

**Mine the cleaned branch, not `main`.** The content-quality audit found fabricated CLI flags in ~44 of 61 modules, concentrated in the generated "enrichment tail". Branch `audit/content-quality-p0-p1` (PR #1) already removed that tail — ~11,000 lines — so it is the higher-signal source. Modules 57 (acquisition) and 58 (eventlog) were rated the cleanest and are worth reading first.

**Candidates are never trusted.** Every mined invocation still has to pass the Phase 3 linter against a real capture. Mining saves the *thinking*, not the *verification* — a mined command with an invented flag must fail exactly as loudly as one written from scratch.

## Phase 2 — Capture pipeline ○

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`scripts/capture_help.py` : for each in-scope tool, run `--help / -h / man` in the right container, store raw output under `capture/<tool>.help.txt`.

- Containers: `cyberlab-aio:v1` and `dfir-aio:v4` on rick (Docker active, 830 GB free).
- **Always timeout-guard** — `vol` once pinned 99% CPU indefinitely on a synthetic image. Hard per-tool timeout, failure recorded not retried blindly.
- Emit a coverage report: which kit tools are covered by a container, which need a booted VM (Kali, FLARE-VM Windows guests), which are GUI-only and have no CLI surface to document.

**Reality check:** the 1,006-tool list includes GUI-only tools (Autopsy, Wireshark GUI, PE-bear) and appliance services (Elasticsearch, Kibana) with no meaningful command surface. Expect the genuinely CLI-documentable set to be substantially smaller. Phase 2's report turns that from a guess into a number before any page is written.

## Phase 3 — Page generation + linter ○

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- Generate a page skeleton per tool from its capture: header, synopsis, complete options table. Mechanical, deterministic — no model invention.
- Human/model adds only the *judgement* columns: purpose, "when you'd use it", invocations, gotchas.
- `scripts/lint.py` gates the build:
- every documented flag exists in the capture;
- every captured flag appears in the page (**completeness**, the hard requirement);
- every tool is in `kit-tools.json`;
- format rules hold (imperative mood, ≤8 examples, placeholder syntax).

## Phase 4 — Print render ○

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Markdown → PDF for field carry. Build artifact only, never hand-edited. Per-capability booklets rather than one 1,000-page brick.

## Phase 5 — Prioritised content ○

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Do not attempt all tools at once. Order by field value:

1. Core forensics: TSK suite, Volatility, Plaso, RegRipper, bulk\_extractor
2. Network: Zeek, Suricata, tshark, tcpdump
3. Malware triage: YARA, ClamAV, capa, FLOSS, Detect-It-Easy
4. Live response / hunting: Velociraptor, osquery, Hayabusa, Chainsaw
5. Offensive-side as needed for exercises

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## Risks

| Risk                                                                                                | Mitigation                                                           |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Fabricated flags (killed the predecessor project)                                                   | Capture-or-it-does-not-ship, enforced by linter                      |
| Scale — 1,006 tools is a lot of pages                                                               | Phase 2 coverage report right-sizes it; Phase 5 prioritises          |
| Kit VMs powered down ( <code>kali</code> , <code>range-linux-web</code> , <code>redinfra01</code> ) | Containers cover most Linux tooling; boot VMs only for the remainder |
| Version drift between capture and deployed kit                                                      | Version + capture date in every page header                          |
| GUI-only tools with no CLI                                                                          | Identify in Phase 2; document workflow, not flags                    |

# Kit Tool List

The binding scope for the quick-reference guide. A tool absent from this list is **not in the kit** and must not be documented in the guide.

Every entry is derived from an upstream machine-readable manifest; none is written from memory. Sources are listed at the end.

## Contents

- [REMnux](#) — 268 tools, 31 categories
- [Kali Linux](#) — 403 tools, 12 categories
- [FLARE-VM](#) — 137 tools, 1 categories
- [SIFT Workstation](#) — 162 tools, 1 categories
- [Security Onion](#) — 36 tools, 9 categories

**Total: 1006 tools across 5 platforms.**

## REMnux

### Analyze Documents / Email Messages

| Tool          | Command(s)               | Purpose                                                                        |
|---------------|--------------------------|--------------------------------------------------------------------------------|
| emldump.py    | —                        | Parse and analyze EML files.                                                   |
| mail-parser   | —                        | Parse raw SMTP and .MSG email messages and generate a parsed object from them. |
| msg-extractor | <code>extract_msg</code> | Extract emails and attachments from MSG files.                                 |
| msgconvert    | —                        | Convert MSG files to MBOX files.                                               |

### Analyze Documents / General

| Tool          | Command(s)             | Purpose                                                                                |
|---------------|------------------------|----------------------------------------------------------------------------------------|
| base64dump.py | —                      | Locate and decode strings encoded in Base64 and other common encodings.                |
| Tesseract OCR | <code>tesseract</code> | Examine images to identify and extract text using optical character recognition (OCR). |

### Analyze Documents / Microsoft Office

| Tool       | Command(s)                                                                                                       | Purpose                                                      |
|------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| EvilClippy | —                                                                                                                | Modify aspects of Microsoft Office documents.                |
| Hachoir    | <code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code> | View, edit, and carve contents of various binary file types. |
| libolecf   | <code>olecfexport</code> , <code>olecfinfo</code> , <code>olecfmount</code> , <code>etc</code>                   | Microsoft Office OLE2 compound documents.                    |

| Tool                 | Command(s)                                                                                                                                                                                                                                                                                             | Purpose                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| msoffcrypto-crack.py | —                                                                                                                                                                                                                                                                                                      | Recover the password of an encrypted Microsoft Office document.                                                 |
| msoffcrypto-tool     | —                                                                                                                                                                                                                                                                                                      | Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key. |
| msoffice-crypt       | —                                                                                                                                                                                                                                                                                                      | Encrypt and decrypt OOXML Microsoft Office documents.                                                           |
| oledump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze OLE2 Structured Storage files.                                                                          |
| olefile              | —                                                                                                                                                                                                                                                                                                      | Python package to parse, read and write MS OLE2 files.                                                          |
| oletools             | <code>mraptor</code> , <code>msodde</code> , <code>olebrowse</code> , <code>oledir</code> , <code>oleid</code> , <code>olemap</code> , <code>olemeta</code> , <code>oleobj</code> , <code>oletimes</code> , <code>olevba</code> , <code>pyxswf</code> , <code>rtfobj</code> , <code>ezhexviewer</code> | Microsoft Office OLE2 compound documents.                                                                       |
| onedump.py           | —                                                                                                                                                                                                                                                                                                      | Extract and analyze embedded files from OneNote documents.                                                      |
| pcode2code           | —                                                                                                                                                                                                                                                                                                      | Decompile VBA macro p-code from Microsoft Office documents.                                                     |
| pcodedmp             | —                                                                                                                                                                                                                                                                                                      | Disassemble VBA p-code.                                                                                         |
| rtfdump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze a suspicious RTF file.                                                                                  |
| SSView               | <code>ssview</code>                                                                                                                                                                                                                                                                                    | Analyze OLE2 Structured Storage files.                                                                          |
| XLMMacroDeobfuscator | <code>xlmddeobfuscator</code> , <code>runxlrd2.py</code>                                                                                                                                                                                                                                               | Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.                            |
| xmldump.py           | —                                                                                                                                                                                                                                                                                                      | Extract contents of XML files, in particular OOXML-formatted Microsoft Office documents.                        |
| zipdump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze zip-compressed files.                                                                                   |

## Analyze Documents / Pdf

| Tool          | Command(s)                                                                                                 | Purpose                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Origamind     | <code>pdfcop</code> , <code>pdfdecompress</code> , <code>pdfdecrypt</code> , <code>pdfextract</code> , etc | Parse, modify, generate PDF files.                                 |
| pdf-parser.py | —                                                                                                          | Examine elements of the PDF file.                                  |
| pdfid.py      | —                                                                                                          | Identify suspicious elements of the PDF file.                      |
| pdfresurrect  | —                                                                                                          | Extract previous versions of content from PDF files.               |
| pdftk-java    | <code>pdftk</code>                                                                                         | Edit, create, and examine PDF files.                               |
| pdftool.py    | —                                                                                                          | Analyze PDF files to identify incremental updates to the document. |
| peepdf-3      | —                                                                                                          | Examine elements of the PDF file.                                  |

| Tool | Command(s) | Purpose                                           |
|------|------------|---------------------------------------------------|
| qpdf | —          | Manipulate (merge, convert, transform) PDF files. |

## Dynamically Reverse-Engineer Code / Elf Files

| Tool                 | Command(s) | Purpose                                                              |
|----------------------|------------|----------------------------------------------------------------------|
| edb                  | —          | An AArch32/x86/x86-64 debugger, well suited for debugging ELF files. |
| GNU Project Debugger | —          | Multi-language debugger.                                             |
| ltrace               | —          | Trace library calls and signals.                                     |
| strace               | —          | Trace process' system calls and signals.                             |

## Dynamically Reverse-Engineer Code / General

| Tool                                  | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Frida                                 | <code>frida</code> , <code>frida-ps</code> , <code>frida-trace</code> , <code>frida-discover</code> , <code>frida-ls-devices</code> , <code>frida-kill</code>        | Trace the execution of a process to analyze its behavior.                                                                |
| r2pipe                                | —                                                                                                                                                                    | Examine binary files, including disassembling and debugging.                                                             |
| radare2                               | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (A |
| Wine                                  | <code>wine</code>                                                                                                                                                    | Run Windows applications.                                                                                                |
| x64dbg Automate MCP (OpenCode skills) | —                                                                                                                                                                    | Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh |

## Dynamically Reverse-Engineer Code / Scripts

| Tool                    | Command(s)                                    | Purpose                                                                                                     |
|-------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| box-js                  | <code>box-js</code> , <code>box-export</code> | Analyze suspicious JavaScript scripts.                                                                      |
| JavaScript Deobfuscator | —                                             | Deobfuscate JavaScript by removing common obfuscation techniques such as string arrays and proxy functions. |
| js_unshroud             | —                                             | Monitor and deobfuscate JavaScript behavior in a headless browser to analyze malicious web pages.           |
| JStillery               | <code>jstillery</code>                        | Deobfuscate JavaScript scripts using AST and Partial Evaluation techniques.                                 |
| objects.js              | —                                             | Emulate common browser and PDF viewer objects, methods, and properties when deobfuscating JavaScript.       |
| PowerShell Core         | <code>pwsh</code>                             | Run PowerShell scripts and commands.                                                                        |
| Rhino Debugger          | <code>rhino-debugger</code>                   | GUI JavaScript debugger.                                                                                    |
| SpiderMonkey            | <code>js</code>                               | Execute and deobfuscate JavaScript using Mozilla's standalone JavaScript engine.                            |

| Tool                   | Command(s)                                   | Purpose                                                                                               |
|------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| SpiderMonkey (Patched) | <code>js-ascii</code> , <code>js-file</code> | Execute and deobfuscate JavaScript using a patched version of Mozilla's standalone JavaScript engine. |
| STPyV8                 | —                                            | Python3 and JavaScript interop engine, fork of the original PyV8 project.                             |
| Webcrack               | —                                            | Deobfuscate, unminify, and unpack bundled JavaScript, including scripts protected with obfuscator.io. |

## Dynamically Reverse-Engineer Code / Shellcode

| Tool              | Command(s)             | Purpose                                                                                         |
|-------------------|------------------------|-------------------------------------------------------------------------------------------------|
| libemu            | —                      | A library for x86 code emulation and shellcode detection.                                       |
| Qiling            | —                      | Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms. |
| runsc             | —                      | Run shellcode to trace and analyze its execution.                                               |
| scdbg             | —                      | Analyze shellcode by emulating its execution.                                                   |
| shcode2exe        | —                      | Convert 32 and 64-bit shellcode to a Windows executable file.                                   |
| shellcode2exe.bat | —                      | Convert 32 and 64-bit shellcode to a Windows executable file.                                   |
| Speakeasy         | —                      | Emulate code execution, including shellcode, Windows drivers, and Windows PE files.             |
| XORSearch         | <code>xorsearch</code> | Locate and decode strings obfuscated using common techniques.                                   |

## Examine Static Properties / .Net

| Tool       | Command(s) | Purpose                                                 |
|------------|------------|---------------------------------------------------------|
| dnfile     | —          | Analyze static properties of .NET files.                |
| dotnetfile | —          | Analyze static properties of .NET files.                |
| monodis    | —          | Disassemble and extract resources from .NET assemblies. |

## Examine Static Properties / Deobfuscation

| Tool                                                    | Command(s)                                                                                    | Purpose                                                                            |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1768.py                                                 | —                                                                                             | Analyze Cobalt Strike beacons.                                                     |
| Balbuzard                                               | <code>balbuzard</code> , <code>bbcrack</code> , <code>bbharvest</code> , <code>bbtrans</code> | Extract and deobfuscate patterns from suspicious files.                            |
| base64dump.py                                           | —                                                                                             | Locate and decode strings encoded in Base64 and other common encodings.            |
| brxor.py                                                | —                                                                                             | Bruteforce XOR'ed strings to find those that are English words.                    |
| Chepy                                                   | <code>chepy</code>                                                                            | Decode and otherwise analyze data using this command-line tool and Python library. |
| Cobalt Strike Configuration Extractor (CSCE) and Parser | <code>csce</code> , <code>list-cs-settings</code>                                             | Analyze Cobalt Strike beacons.                                                     |
| cs-analyze-processdump.py                               | —                                                                                             | Analyze Cobalt Strike beacon process dumps to detect sleep mask encoding.          |

| Tool                   | Command(s)             | Purpose                                                                            |
|------------------------|------------------------|------------------------------------------------------------------------------------|
| cs-decrypt-metadata.py | —                      | Decrypt Cobalt Strike metadata.                                                    |
| cs-extract-key.py      | —                      | Extract AES and HMAC keys from Cobalt Strike beacon process memory.                |
| cut-bytes.py           | —                      | Cut out a part of a data stream.                                                   |
| CyberChef              | <code>cyberchef</code> | Decode and otherwise analyze data using this browser app.                          |
| DC3-MWCP               | <code>mwcp</code>      | Parsing configuration information from malware.                                    |
| ex_pe_xor.py           | —                      | Search an XOR'ed file for indications of executable binaries.                      |
| FLOSS                  | <code>floss</code>     | Extract and deobfuscate strings from PE executables.                               |
| format-bytes.py        | —                      | Decompose structured binary data with format strings.                              |
| hex-to-bin.py          | —                      | Convert hexadecimal text dumps to binary data.                                     |
| Malchive               | —                      | Perform static analysis of various aspects of malicious code.                      |
| NoMoreXOR.py           | —                      | Help guess a file's 256-byte XOR by using frequency analysis.                      |
| numbers-to-string.py   | —                      | Translate number sequences into ASCII characters.                                  |
| re-search.py           | —                      | Search files using regular expressions from a built-in library or custom patterns. |
| sets.py                | —                      | Perform set operations on lines or bytes in text files.                            |
| strdeob.pl             | —                      | Locate and decode stack strings in executable files.                               |
| translate.py           | —                      | Translate bytes according to a Python expression.                                  |
| unicode                | —                      | Display Unicode character properties.                                              |
| unXOR                  | —                      | Deobfuscate XOR'ed files.                                                          |
| xor-kpa.py             | —                      | Implement a XOR known plaintext attack.                                            |
| xorBruteForcer.py      | —                      | Bruteforce an XOR-encoded file.                                                    |
| XORSearch              | <code>xorsearch</code> | Locate and decode strings obfuscated using common techniques.                      |
| xorsearch.py           | —                      | Search for XOR, ROL, ROT, and SHIFT encoded strings with YARA and regex support.   |
| XORStrings             | —                      | Search for XOR encoded strings in a file.                                          |
| xortool                | —                      | Analyze XOR-encoded data.                                                          |

## Examine Static Properties / Elf Files

| Tool       | Command(s)              | Purpose                                                                             |
|------------|-------------------------|-------------------------------------------------------------------------------------|
| pyelftools | <code>readelf.py</code> | Python library for parsing and analyzing ELF files and DWARF debugging information. |



## Examine Static Properties / General

| Tool                 | Command(s)                                                                                                          | Purpose                                                                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 7-Zip                | <code>7za</code>                                                                                                    | Compress and decompress files using a variety of algorithms.                           |
| binwalk              | —                                                                                                                   | Extract and analyze firmware images.                                                   |
| bulk_extractor       | —                                                                                                                   | Extract interesting strings from binary files.                                         |
| ClamAV               | <code>clamscan</code> , <code>freshclam</code>                                                                      | Scan files for malware signatures.                                                     |
| Detect-It-Easy       | —                                                                                                                   | Determine types of files and examine file properties.                                  |
| disitool             | <code>disitool.py</code>                                                                                            | Manipulate embedded digital signatures.                                                |
| DroidLysis           | <code>droidlysis</code>                                                                                             | Perform static analysis of Android applications.                                       |
| ExifTool             | <code>exiftool</code>                                                                                               | Tool to read from, write to, and edit EXIF metadata of various file types.             |
| file                 | —                                                                                                                   | Identify file type using "magic" numbers.                                              |
| file-magic.py        | —                                                                                                                   | Identify file types using the Python magic module.                                     |
| Hachoir              | <code>hachoir-grep</code> , <code>hachoir-metadata</code> ,<br><code>hachoir-strip</code> , <code>hachoir-wx</code> | View, edit, and carve contents of various binary file types.                           |
| Hash ID              | <code>hash-id.py</code>                                                                                             | Identify different types of hashes.                                                    |
| LIEF                 | —                                                                                                                   | Parse and analyze PE, ELF, MachO, DEX, OAT, VDEX, ART, and DWARF executable formats.   |
| Magika               | —                                                                                                                   | Identify file type using signatures.                                                   |
| Malcat Lite          | —                                                                                                                   | Analyze binary files using a hex editor, disassembler, and file dissector.             |
| msitools             | —                                                                                                                   | Create, inspect and extract Windows Installer (.msi) files.                            |
| Name-That-Hash       | <code>nth</code>                                                                                                    | Identify different types of hashes.                                                    |
| numbers-to-string.py | —                                                                                                                   | Convert decimal numbers to strings.                                                    |
| re-search.py         | —                                                                                                                   | Search the file for built-in regular expressions of common suspicious artifacts.       |
| signsrch             | —                                                                                                                   | Find patterns of common encryption, compression, or encoding algorithms.               |
| Sleuth Kit           | —                                                                                                                   | Analyze disk images and recover files from them.                                       |
| ssdeep               | —                                                                                                                   | Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.         |
| strings.py           | —                                                                                                                   | Extract ASCII and Unicode strings from binary files with length sorting and filtering. |
| thefuzz              | —                                                                                                                   | Fuzzy String Matching in Python.                                                       |
| TrID                 | <code>trid</code> , <code>tridupdate</code>                                                                         | Identify file type using signatures.                                                   |
| wxHexEditor          | —                                                                                                                   | Hex editor.                                                                            |

| Tool             | Command(s) | Purpose                                                                                                        |
|------------------|------------|----------------------------------------------------------------------------------------------------------------|
| Yara Rules       | —          | Scan a file with YARA rules to identify capabilities and behaviors (packer detection, anti-debug, networking). |
| YARA-Forge Rules | —          | Scan files with curated YARA rules from 45+ sources for malware family identification.                         |

## Examine Static Properties / Go

| Tool    | Command(s)           | Purpose                                                                                               |
|---------|----------------------|-------------------------------------------------------------------------------------------------------|
| GoReSym | <code>GoReSym</code> | Extract metadata and symbols from Go binaries, including stripped ones.                               |
| Redress | <code>redress</code> | Analyze stripped Go binaries to recover symbols, types, source structure, and integrate with Radare2. |

## Examine Static Properties / Pe Files

| Tool                  | Command(s)                                                                                                                                                                                                                                   | Purpose                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| bearparser            | <code>bearcommander</code>                                                                                                                                                                                                                   | Parse PE file contents.                                                     |
| debloat               | —                                                                                                                                                                                                                                            | Remove junk contents from bloated Windows executables.                      |
| disitool.py           | —                                                                                                                                                                                                                                            | Extract, delete, copy, and inject digital signatures in PE files.           |
| dllcharacteristics.py | —                                                                                                                                                                                                                                            | Read and set DLL characteristics of a PE file.                              |
| Manalyze              | —                                                                                                                                                                                                                                            | Perform static analysis of suspicious PE files.                             |
| PE Tree               | <code>pe-tree</code>                                                                                                                                                                                                                         | Examine contents and structure of PE files.                                 |
| pecheck.py            | —                                                                                                                                                                                                                                            | Analyze static properties of PE files.                                      |
| pedump                | —                                                                                                                                                                                                                                            | Statically analyze PE files and extract their components (e.g., resources). |
| pefile                | —                                                                                                                                                                                                                                            | Python library for analyzing static properties of PE files.                 |
| PEframe               | <code>peframe</code>                                                                                                                                                                                                                         | Statically analyze PE and Microsoft Office files.                           |
| pev                   | <code>pestr</code> , <code>readpe</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>peldd</code> , <code>peres</code>                                                                                          | Analyze PE files and extract strings from them.                             |
| PortEx                | <code>portex</code>                                                                                                                                                                                                                          | Statically analyze PE files.                                                |
| readpe (formerly pev) | <code>readpe</code> , <code>pestr</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>pesec</code> , <code>peldd</code> , <code>pepack</code> , <code>peres</code> , <code>ofs2rva</code> , <code>rva2ofs</code> | Analyze PE files and extract strings from them.                             |

## Explore Network Interactions / Connecting

| Tool            | Command(s)           | Purpose                                                                                                             |
|-----------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Anomy           | <code>anomy</code>   | A wrapper around wget, ssh, sftp, ftp, and telnet to route these connections through Tor to anonymize your traffic. |
| cURL            | <code>curl</code>    | Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool. |
| EPIC IRC Client | <code>epic5</code>   | Examine IRC activities with this IRC client.                                                                        |
| GNU Wget        | <code>wget</code>    | Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.                                  |
| netcat          | <code>nc</code>      | Read and write data across network connections.                                                                     |
| thug            | —                    | Examine suspicious website using this low-interaction honeyclient.                                                  |
| tor             | —                    | Obfuscate your origins by routing traffic through a network of anonymizing nodes.                                   |
| Unfurl          | —                    | Deconstruct and decode data from a URL.                                                                             |
| zbarimg         | <code>zbarimg</code> | Decode QR codes and barcodes from image files.                                                                      |

## Explore Network Interactions / Monitoring

| Tool                         | Command(s)                                                            | Purpose                                                                                                                  |
|------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Burp Suite Community Edition | <code>burpsuite</code>                                                | Investigate website interactions using this web proxy.                                                                   |
| cs-parse-traffic.py          | —                                                                     | Decrypt and parse Cobalt Strike beacon network traffic.                                                                  |
| mitmproxy                    | <code>mitmproxy</code> , <code>mitmdump</code> , <code>mitmweb</code> | Investigate website interactions using this web proxy.                                                                   |
| monitor-network              | —                                                                     | Monitor traffic on the first active network interface using tshark, printing a live summary to the screen or saving it t |
| Network Miner Free Edition   | <code>networkminer</code>                                             | Examine network traffic and carve PCAP capture files.                                                                    |
| ngrep                        | —                                                                     | Look for patterns in network traffic.                                                                                    |
| PolarProxy                   | <code>polarproxy</code>                                               | Intercept and decrypt TLS traffic.                                                                                       |
| tcpdump                      | —                                                                     | Capture and analyze network traffic with this command-line sniffer.                                                      |
| tcpflow                      | —                                                                     | Analyze the flow of network traffic.                                                                                     |
| tcpick                       | —                                                                     | Capture and analyze network traffic with this command-line sniffer.                                                      |
| tcpxtract                    | —                                                                     | Extract files from network traffic.                                                                                      |
| tshark                       | —                                                                     | Capture and analyze network traffic with this console-based sniffer.                                                     |
| wireshark                    | —                                                                     | Capture and analyze network traffic with this sniffer.                                                                   |

## Explore Network Interactions / Services

| Tool           | Command(s) | Purpose                                                                                            |
|----------------|------------|----------------------------------------------------------------------------------------------------|
| accept-all-ips | —          | Accept connections to all IPv4 and IPv6 addresses and redirect it to the corresponding local port. |
| dnsresolver.py | —          | DNS resolver tool for dynamic analysis with wildcard and tracking support.                         |

| Tool       | Command(s)           | Purpose                                                               |
|------------|----------------------|-----------------------------------------------------------------------|
| fakedns    | —                    | Respond to DNS queries with the specified IP address.                 |
| fakemail   | —                    | Intercept and examine SMTP email activity with this fake SMTP server. |
| FakeNet-NG | —                    | Emulate common network services and interact with malware.            |
| INetSim    | <code>inetsim</code> | Emulate common network services and interact with malware.            |
| inspircd 3 | —                    | Examine IRC activity with this IRC server.                            |
| netcat     | <code>nc</code>      | Read and write data across network connections.                       |
| Nginx      | —                    | Web server.                                                           |

## Gather And Analyze Data / General

| Tool              | Command(s)                        | Purpose                                                                                           |
|-------------------|-----------------------------------|---------------------------------------------------------------------------------------------------|
| DeXRAY            | <code>dexray</code>               | Extract and decode data from antivirus quarantine files.                                          |
| dissect           | —                                 | Perform a variety of forensics and incident response tasks using this DFIR framework and toolset. |
| dnslib            | —                                 | Python library to encode/decode DNS wire-format packets.                                          |
| ioc_parser        | —                                 | Extract IOCs from security report PDFs.                                                           |
| ipwhois           | —                                 | Retrieve and parse whois data for IP addresses.                                                   |
| malwoverview      | <code>malwoverview</code>         | Query public repositories of malware data (e.g., VirusTotal, HybridAnalysis).                     |
| nsrlookup         | —                                 | Look up MD5 file hashes in the NIST National Software Reference Library (NSRL).                   |
| pdnstool          | —                                 | Query passive DNS databases for DNS data.                                                         |
| Scalpel           | —                                 | Carve contents out of binary files, such as partitions.                                           |
| time-decode       | —                                 | Decode and encode date and timestamps.                                                            |
| virustotal-search | <code>virustotal-search.py</code> | Search VirusTotal for file hashes.                                                                |
| virustotal-submit | <code>virustotal-submit.py</code> | Submit files to VirusTotal.                                                                       |
| Yara              | <code>yara</code>                 | Identify and classify malware samples using Yara rules.                                           |
| YARA-X            | —                                 | Scan files using YARA rules, the next generation of YARA written in Rust.                         |

## General Utilities / General

| Tool       | Command(s)           | Purpose                                                                                                             |
|------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| 7-Zip      | <code>7za</code>     | Compress and decompress files using a variety of algorithms.                                                        |
| cabextract | —                    | Extract Microsoft cabinet (cab) files.                                                                              |
| cURL       | <code>curl</code>    | Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool. |
| Docker     | —                    | Run and manage containers.                                                                                          |
| Firefox    | <code>firefox</code> | Web browser.                                                                                                        |

| Tool             | Command(s)                                                 | Purpose                                                                                                                  |
|------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| GNOME Calculator | <code>galculator</code>                                    | Calculator.                                                                                                              |
| IBus             | <code>ibus-setup</code>                                    | Adjust input methods for the GUI.                                                                                        |
| Info-ZIP         | <code>zip</code> , <code>unzip</code>                      | Compress and decompress files using the zip algorithm.                                                                   |
| myip             | —                                                          | Determine the IP address of the default network interface.                                                               |
| myjson-filter.py | —                                                          | Filter data formatted using the JSON format used by Didier Stevens' tools.                                               |
| nasm             | —                                                          | An x86-64 assembler.                                                                                                     |
| Nautilus         | —                                                          | Graphical file manager.                                                                                                  |
| OpenSSH          | <code>sftp</code> , <code>ssh</code> ,<br><code>etc</code> | Initiate and receive SSH and SFTP connections.                                                                           |
| PowerShell Core  | <code>pwsh</code>                                          | Run PowerShell scripts and commands.                                                                                     |
| RAR              | <code>rar</code>                                           | Compress and decompress files using a variety of algorithms.                                                             |
| REMnux Installer | —                                                          | Install and update the REMnux distro.                                                                                    |
| restrict-egress  | —                                                          | Restrict outbound network access to an allowlist of domains and CIDRs using an nftables default-deny egress policy. It i |
| sortcanon.py     | —                                                          | Sort text files using canonicalization functions built into this tool.                                                   |
| SQLite           | <code>sqlite3</code>                                       | Manage and interact with SQL database files.                                                                             |
| sshpas           | <code>sshpas</code>                                        | Supply a password to SSH non-interactively for automated logins.                                                         |
| texteditor.py    | —                                                          | Edit text files from the command line using search-and-replace commands.                                                 |
| unrar-free       | <code>unrar</code>                                         | Decompress files using a variety of algorithms.                                                                          |
| Wine             | <code>wine</code>                                          | Run Windows applications.                                                                                                |

## Investigate System Interactions / General

| Tool                   | Command(s)           | Purpose                                                                            |
|------------------------|----------------------|------------------------------------------------------------------------------------|
| ProcDOT                | <code>procdot</code> | Visualize and examine the output of Process Monitor.                               |
| ProcmonMCP             | —                    | MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures. |
| sandfly-processdecloak | —                    | Find hidden processes on the local Linux system.                                   |
| Unhide                 | —                    | Find hidden processes or connections on the local Linux system.                    |

## Perform Memory Forensics / General

| Tool                 | Command(s)              | Purpose                                              |
|----------------------|-------------------------|------------------------------------------------------|
| AESKeyFinder         | <code>aeskeyfind</code> | Find 128-bit and 256-bit AES keys in a memory image. |
| bulk_extractor       | —                       | Extract interesting strings from binary files.       |
| RSAKeyFinder         | <code>rsakeyfind</code> | Find BER-encoded RSA private keys in a memory image. |
| Volatility Framework | —                       | Memory forensics tool and framework.                 |

## Statically Analyze Code / .Net

| Tool   | Command(s) | Purpose                               |
|--------|------------|---------------------------------------|
| de4dot | —          | Deobfuscate and unpack .NET programs. |
| ILSpy  | —          | Examine and decompile .NET programs.  |

## Statically Analyze Code / Android

| Tool                  | Command(s)                                        | Purpose                                                                                     |
|-----------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------|
| androguard            | —                                                 | Examine Android files.                                                                      |
| AndroidProjectCreator | —                                                 | Convert an Android APK application file into an Android Studio project for easier analysis. |
| APKiD                 | <code>apkid</code>                                | Identify compilers, packers, and obfuscators used to protect Android APK and DEX files.     |
| apktool               | —                                                 | Reverse-engineer Android APK files.                                                         |
| baksmali              | —                                                 | Disassembler for the dex format used by Dalvik, Android's Java VM implementation.           |
| dex2jar               | <code>dex-tools</code> , <code>d2j-dex2jar</code> | Examine Dalvik Executable (dex) files.                                                      |
| DroidLysis            | <code>droidlysis</code>                           | Perform static analysis of Android applications.                                            |
| JADX                  | <code>jadx</code> , <code>jadx-gui</code>         | Generate Java source code from Dalvik Executable (dex) and Android APK files.               |

## Statically Analyze Code / General

| Tool           | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Cutter         | —                                                                                                                                                                    | Reverse engineering platform powered by Rizin.                                                                           |
| Detect-It-Easy | —                                                                                                                                                                    | Determine types of files and examine file properties.                                                                    |
| Ghidra         | —                                                                                                                                                                    | Software reverse engineering tool suite.                                                                                 |
| objdump        | —                                                                                                                                                                    | Disassemble binary files.                                                                                                |
| Qiling         | —                                                                                                                                                                    | Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.                          |
| radare2        | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A |
| Vivisect       | <code>vivbin</code> , <code>vdbbin</code>                                                                                                                            | Statically examine and emulate binary files.                                                                             |

## Statically Analyze Code / Java

| Tool            | Command(s)                 | Purpose                 |
|-----------------|----------------------------|-------------------------|
| cfr             | —                          | Java decompiler.        |
| Java IDX Parser | <code>idx_parser.py</code> | Analyze Java IDX files. |

| Tool                   | Command(s)           | Purpose                                    |
|------------------------|----------------------|--------------------------------------------|
| Javassist              | —                    | Java bytecode engineering toolkit/library. |
| JD-GUI Java Decompiler | <code>jd-gui</code>  | Java decompiler with GUI.                  |
| Procyon                | <code>procyon</code> | Java decompiler.                           |

## Statically Analyze Code / Pe-Files

| Tool                                  | Command(s)              | Purpose                                                                             |
|---------------------------------------|-------------------------|-------------------------------------------------------------------------------------|
| binnee (Binary Emulation Environment) | —                       | Analyze I/O operations of a suspicious PE file by emulating its execution.          |
| capa                                  | —                       | Detect suspicious capabilities in PE files.                                         |
| Malchive                              | —                       | Perform static analysis of various aspects of malicious code.                       |
| mbcscan                               | <code>mbcscan.py</code> | Scan a PE file to list the associated Malware Behavior Catalog (MBC) details.       |
| Speakeasy                             | —                       | Emulate code execution, including shellcode, Windows drivers, and Windows PE files. |

## Statically Analyze Code / Python

| Tool                  | Command(s)                               | Purpose                                                                                            |
|-----------------------|------------------------------------------|----------------------------------------------------------------------------------------------------|
| Decompyle++           | <code>pycdas</code> , <code>pycdc</code> | Python bytecode disassembler and decompiler.                                                       |
| PyInstaller Extractor | <code>pyinstxtractor.py</code>           | Extract contents of a PyInstaller-generated PE files.                                              |
| pyinstxtractor-ng     | <code>pyinstxtractor-ng</code>           | Extract contents of PyInstaller-generated executables without requiring a matching Python version. |
| uncompyle6            | <code>uncompyle6</code>                  | Python cross-version bytecode decompiler for Python 1.0 through 3.8.                               |

## Statically Analyze Code / Scripts

| Tool                      | Command(s)                 | Purpose                                                                                                         |
|---------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Autolt-Ripper             | <code>autoit-ripper</code> | Extract Autolt scripts embedded in PE binaries.                                                                 |
| decode-vbe.py             | —                          | Decode encoded VBS scripts (VBE).                                                                               |
| GootLoaderAutoJsDecode.py | —                          | Statically deobfuscate GootLoader (GOOTLOADER) malicious JScript to recover the payload and extract C2 domains. |
| JS Beautifier             | <code>js-beautify</code>   | Reformat JavaScript scripts for easier analysis.                                                                |

## Statically Analyze Code / Unpacking

| Tool     | Command(s)                                     | Purpose                                                                         |
|----------|------------------------------------------------|---------------------------------------------------------------------------------|
| binwalk  | —                                              | Extract and analyze firmware images.                                            |
| Bytehist | <code>bytehist</code>                          | Generate byte-usage-histograms for all types of files with a focus on PE files. |
| ClamAV   | <code>clamscan</code> , <code>freshclam</code> | Scan files for malware signatures.                                              |

| Tool | Command(s)                                  | Purpose                              |
|------|---------------------------------------------|--------------------------------------|
| TrID | <code>trid</code> , <code>tridupdate</code> | Identify file type using signatures. |
| UPX  | <code>upx</code>                            | Pack and unpack PE files.            |

## Use Artificial Intelligence / General

| Tool                                  | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| GhidraAssistMCP                       | —                                                                                                                                                                    | MCP server for AI-assisted reverse engineering in Ghidra.                                                                |
| OpenCode                              | —                                                                                                                                                                    | Open-source AI coding agent for the terminal.                                                                            |
| ProcmonMCP                            | —                                                                                                                                                                    | MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.                                       |
| radare2                               | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A |
| REMnux MCP Server                     | <code>remnux-mcp-server</code>                                                                                                                                       | MCP server for using the REMnux malware analysis toolkit via AI assistants.                                              |
| x64dbg Automate MCP (OpenCode skills) | —                                                                                                                                                                    | Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh |

## View Or Edit Files / General

| Tool               | Command(s)                                                                                    | Purpose                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| dos2unix           | <code>dos2unix</code> , <code>mac2unix</code> , <code>unix2dos</code> , <code>unix2mac</code> | Convert text files with Windows or macOS line breaks to Unix line breaks and vice versa.         |
| Evince             | <code>evince</code>                                                                           | View documents in a variety of formats, including PDF.                                           |
| feh                | —                                                                                             | View images.                                                                                     |
| ImageMagick        | <code>display</code> , <code>convert</code> , <code>mogrify</code> , etc                      | View and manipulate image and related files.                                                     |
| Mermaid Viewer     | —                                                                                             | View Mermaid diagrams, such as AI-generated code-analysis workflow diagrams, in a local browser. |
| SciTE              | —                                                                                             | Edit text files.                                                                                 |
| VBinDiff           | <code>vbindiff</code>                                                                         | Compare binary files.                                                                            |
| Visual Studio Code | <code>code</code>                                                                             | Powerful source code editor.                                                                     |
| wxHexEditor        | —                                                                                             | Hex editor.                                                                                      |

## Kali Linux

### Crypto Stego

| Tool   | Command(s)          | Purpose |
|--------|---------------------|---------|
| aesfix | <code>aesfix</code> |         |



| Tool       | Command(s) | Purpose |
|------------|------------|---------|
| aeskeyfind | aeskeyfind |         |
| ccrypt     | ccrypt     |         |
| steghide   | steghide   |         |
| stegosuite | stegosuite |         |
| stegsnow   | stegsnow   |         |

## Database

| Tool           | Command(s)     | Purpose |
|----------------|----------------|---------|
| jsql-injection | jsql-injection |         |
| mdbtools       | mdbtools       |         |
| oscanner       | oscanner       |         |
| sidguesser     | sidguesser     |         |
| sqldict        | sqldict        |         |
| sqlitebrowser  | sqlitebrowser  |         |
| sqlmap         | sqlmap         |         |
| sqlninja       | sqlninja       |         |
| sqlsus         | sqlsus         |         |
| tnscmd10g      | tnscmd10g      |         |

## Exploitation

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| armitage             | armitage             |         |
| beef-xss             | beef-xss             |         |
| exploitdb            | exploitdb            |         |
| metasploit-framework | metasploit-framework |         |
| msfpc                | msfpc                |         |
| set                  | set                  |         |
| shellnoob            | shellnoob            |         |
| sqlmap               | sqlmap               |         |
| termineter           | termineter           |         |

## Forensics

| Tool         | Command(s)   | Purpose |
|--------------|--------------|---------|
| 7zip         | 7zip         |         |
| afflib-tools | afflib-tools |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| apktool            | apktool            |         |
| autopsy            | autopsy            |         |
| binwalk            | binwalk            |         |
| binwalk3           | binwalk3           |         |
| bulk-extractor     | bulk-extractor     |         |
| bytecode-viewer    | bytecode-viewer    |         |
| cabextract         | cabextract         |         |
| chkrootkit         | chkrootkit         |         |
| creddump7          | creddump7          |         |
| dc3dd              | dc3dd              |         |
| dcfldd             | dcfldd             |         |
| ddrescue           | ddrescue           |         |
| dumpzilla          | dumpzilla          |         |
| edb-debugger       | edb-debugger       |         |
| ewf-tools          | ewf-tools          |         |
| exifprobe          | exifprobe          |         |
| exiv2              | exiv2              |         |
| ext3grep           | ext3grep           |         |
| ext4magic          | ext4magic          |         |
| extundelete        | extundelete        |         |
| fcrackzip          | fcrackzip          |         |
| firmware-mod-kit   | firmware-mod-kit   |         |
| foremost           | foremost           |         |
| forensic-artifacts | forensic-artifacts |         |
| forensics-colorize | forensics-colorize |         |
| galleta            | galleta            |         |
| gdb                | gdb                |         |
| gpart              | gpart              |         |
| gparted            | gparted            |         |
| grokvt             | grokvt             |         |
| guymager           | guymager           |         |
| hashdeep           | hashdeep           |         |
| inetsim            | inetsim            |         |
| jadx               | jadx               |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| jasvasnoop         | jasvasnoop         |         |
| libhivex-bin       | libhivex-bin       |         |
| libsmali-java      | libsmali-java      |         |
| lvm2               | lvm2               |         |
| lynis              | lynis              |         |
| mac-robber         | mac-robber         |         |
| magicrescue        | magicrescue        |         |
| md5deep            | md5deep            |         |
| mdbtools           | mdbtools           |         |
| memdump            | memdump            |         |
| metacam            | metacam            |         |
| missidentify       | missidentify       |         |
| myrescue           | myrescue           |         |
| nasm               | nasm               |         |
| nasty              | nasty              |         |
| ollydbg            | ollydbg            |         |
| parted             | parted             |         |
| pasco              | pasco              |         |
| pdf-parser         | pdf-parser         |         |
| pdfid              | pdfid              |         |
| plaso              | plaso              |         |
| polenum            | polenum            |         |
| pst-utils          | pst-utils          |         |
| python3-capstone   | python3-capstone   |         |
| python3-dfdatetime | python3-dfdatetime |         |
| python3-dfvfs      | python3-dfvfs      |         |
| python3-dfwinreg   | python3-dfwinreg   |         |
| python3-distorm3   | python3-distorm3   |         |
| radare2            | radare2            |         |
| readpe             | readpe             |         |
| recoverdm          | recoverdm          |         |
| recoverjpeg        | recoverjpeg        |         |
| recstudio          | recstudio          |         |
| reglookup          | reglookup          |         |

| Tool          | Command(s)    | Purpose |
|---------------|---------------|---------|
| regripper     | regripper     |         |
| rephrase      | rephrase      |         |
| rifiuti       | rifiuti       |         |
| rifiuti2      | rifiuti2      |         |
| rizin-cutter  | rizin-cutter  |         |
| rkhunter      | rkhunter      |         |
| rsakeyfind    | rsakeyfind    |         |
| rz-ghidra     | rz-ghidra     |         |
| safecopy      | safecopy      |         |
| samdump2      | samdump2      |         |
| scalpel       | scalpel       |         |
| scrounge-ntfs | scrounge-ntfs |         |
| sleuthkit     | sleuthkit     |         |
| sqlitebrowser | sqlitebrowser |         |
| ssdeep        | ssdeep        |         |
| tcpdump       | tcpdump       |         |
| tcpflow       | tcpflow       |         |
| tcpick        | tcpick        |         |
| tcpreplay     | tcpreplay     |         |
| truecrack     | truecrack     |         |
| undbx         | undbx         |         |
| unhide        | unhide        |         |
| unrar         | unrar         |         |
| upx-ucl       | upx-ucl       |         |
| vinetto       | vinetto       |         |
| wce           | wce           |         |
| winregfs      | winregfs      |         |
| wireshark     | wireshark     |         |
| xmount        | xmount        |         |
| xplico        | xplico        |         |
| yara          | yara          |         |

## Fuzzing

| Tool  | Command(s) | Purpose |
|-------|------------|---------|
| afl++ | afl++      |         |
| sfuzz | sfuzz      |         |
| spike | spike      |         |
| wfuzz | wfuzz      |         |

## Information Gathering

| Tool        | Command(s)  | Purpose |
|-------------|-------------|---------|
| Otrace      | Otrace      |         |
| arping      | arping      |         |
| braa        | braa        |         |
| dmitry      | dmitry      |         |
| dnsenum     | dnsenum     |         |
| dnsmap      | dnsmap      |         |
| dnsrecon    | dnsrecon    |         |
| dnstracer   | dnstracer   |         |
| dnswalk     | dnswalk     |         |
| enum4linux  | enum4linux  |         |
| fierce      | fierce      |         |
| firewalk    | firewalk    |         |
| fping       | fping       |         |
| fragrouter  | fragrouter  |         |
| ftester     | ftester     |         |
| hping3      | hping3      |         |
| ike-scan    | ike-scan    |         |
| intrace     | intrace     |         |
| irpas       | irpas       |         |
| lbd         | lbd         |         |
| legion      | legion      |         |
| maltego     | maltego     |         |
| masscan     | masscan     |         |
| metagoofil  | metagoofil  |         |
| nbtscan     | nbtscan     |         |
| ncat        | ncat        |         |
| netdiscover | netdiscover |         |

| Tool           | Command(s)     | Purpose |
|----------------|----------------|---------|
| netmask        | netmask        |         |
| nmap           | nmap           |         |
| onesixtyone    | onesixtyone    |         |
| p0f            | p0f            |         |
| qsslcaudit     | qsslcaudit     |         |
| recon-ng       | recon-ng       |         |
| smbmap         | smbmap         |         |
| smtp-user-enum | smtp-user-enum |         |
| snmpcheck      | snmpcheck      |         |
| ssldump        | ssldump        |         |
| sslh           | sslh           |         |
| sslscan        | sslscan        |         |
| sslyze         | sslyze         |         |
| swaks          | swaks          |         |
| thc-ipv6       | thc-ipv6       |         |
| theharvester   | theharvester   |         |
| tlssled        | tlssled        |         |
| twofi          | twofi          |         |
| unicornscan    | unicornscan    |         |
| urlcrazy       | urlcrazy       |         |
| wafw00f        | wafw00f        |         |
| zenmap         | zenmap         |         |

## Passwords

| Tool                | Command(s)          | Purpose |
|---------------------|---------------------|---------|
| cewl                | cewl                |         |
| chntpw              | chntpw              |         |
| cisco-auditing-tool | cisco-auditing-tool |         |
| cmospwd             | cmospwd             |         |
| crackle             | crackle             |         |
| creddump7           | creddump7           |         |
| crunch              | crunch              |         |
| fcrackzip           | fcrackzip           |         |
| freerdp3-x11        | freerdp3-x11        |         |

| Tool             | Command(s)       | Purpose |
|------------------|------------------|---------|
| gpp-decrypt      | gpp-decrypt      |         |
| hash-identifier  | hash-identifier  |         |
| hashcat          | hashcat          |         |
| hashcat-utils    | hashcat-utils    |         |
| hashid           | hashid           |         |
| hydra            | hydra            |         |
| hydra-gtk        | hydra-gtk        |         |
| john             | john             |         |
| johnny           | johnny           |         |
| maskprocessor    | maskprocessor    |         |
| medusa           | medusa           |         |
| mimikatz         | mimikatz         |         |
| ncrack           | ncrack           |         |
| onesixtyone      | onesixtyone      |         |
| ophcrack         | ophcrack         |         |
| ophcrack-cli     | ophcrack-cli     |         |
| pack             | pack             |         |
| pack2            | pack2            |         |
| passing-the-hash | passing-the-hash |         |
| patator          | patator          |         |
| pdfcrack         | pdfcrack         |         |
| pipal            | pipal            |         |
| polenum          | polenum          |         |
| rainbowcrack     | rainbowcrack     |         |
| rarcrack         | rarcrack         |         |
| rcracki-mt       | rcracki-mt       |         |
| rsmangler        | rsmangler        |         |
| samdump2         | samdump2         |         |
| seclists         | seclists         |         |
| sipcrack         | sipcrack         |         |
| sipvicious       | sipvicious       |         |
| smbmap           | smbmap           |         |
| sqldict          | sqldict          |         |
| statsprocessor   | statsprocessor   |         |

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| sucrack         | sucrack         |         |
| thc-pptp-bruter | thc-pptp-bruter |         |
| truecrack       | truecrack       |         |
| twofi           | twofi           |         |
| wordlists       | wordlists       |         |

## Post Exploitation

| Tool         | Command(s)   | Purpose |
|--------------|--------------|---------|
| cymothoa     | cymothoa     |         |
| dbd          | dbd          |         |
| dns2tcp      | dns2tcp      |         |
| exe2hexbat   | exe2hexbat   |         |
| iodine       | iodine       |         |
| laudanum     | laudanum     |         |
| mimikatz     | mimikatz     |         |
| miredo       | miredo       |         |
| nishang      | nishang      |         |
| powersploit  | powersploit  |         |
| proxychains4 | proxychains4 |         |
| proxytunnel  | proxytunnel  |         |
| ptunnel      | ptunnel      |         |
| pwnat        | pwnat        |         |
| sbd          | sbd          |         |
| shellter     | shellter     |         |
| sslh         | sslh         |         |
| stunnel4     | stunnel4     |         |
| udptunnel    | udptunnel    |         |
| veil         | veil         |         |
| webacoo      | webacoo      |         |
| weevely      | weevely      |         |

## Reverse Engineering

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| apktool         | apktool         |         |
| bytecode-viewer | bytecode-viewer |         |



| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| clang                | clang                |         |
| dex2jar              | dex2jar              |         |
| edb-debugger         | edb-debugger         |         |
| jadx                 | jadx                 |         |
| jasnnoop             | jasnnoop             |         |
| jd-gui               | jd-gui               |         |
| metasploit-framework | metasploit-framework |         |
| ollydbg              | ollydbg              |         |
| radare2              | radare2              |         |
| recstudio            | recstudio            |         |
| rizin                | rizin                |         |
| rizin-cutter         | rizin-cutter         |         |
| rz-ghidra            | rz-ghidra            |         |

## Sniffing Spoofing

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| above              | above              |         |
| bettercap          | bettercap          |         |
| darkstat           | darkstat           |         |
| dnscraf            | dnscraf            |         |
| driftnet           | driftnet           |         |
| dsniff             | dsniff             |         |
| ettercap-graphical | ettercap-graphical |         |
| ferret-sidejack    | ferret-sidejack    |         |
| fiked              | fiked              |         |
| hamster-sidejack   | hamster-sidejack   |         |
| hexinject          | hexinject          |         |
| isr-evilgrade      | isr-evilgrade      |         |
| macchanger         | macchanger         |         |
| mitmproxy          | mitmproxy          |         |
| netsniff-ng        | netsniff-ng        |         |
| rebind             | rebind             |         |
| responder          | responder          |         |
| sniffjoke          | sniffjoke          |         |

| Tool       | Command(s) | Purpose |
|------------|------------|---------|
| sslsniff   | sslsniff   |         |
| sslsplit   | sslsplit   |         |
| tcpflow    | tcpflow    |         |
| tcpreplay  | tcpreplay  |         |
| wifi-honey | wifi-honey |         |
| wireshark  | wireshark  |         |

## Vulnerability

| Tool                   | Command(s)             | Purpose |
|------------------------|------------------------|---------|
| afl++                  | afl++                  |         |
| bed                    | bed                    |         |
| cisco-auditing-tool    | cisco-auditing-tool    |         |
| cisco-global-exploiter | cisco-global-exploiter |         |
| cisco-ocs              | cisco-ocs              |         |
| cisco-torch            | cisco-torch            |         |
| copy-router-config     | copy-router-config     |         |
| dhcpig                 | dhcpig                 |         |
| enumiax                | enumiax                |         |
| gvm                    | gvm                    |         |
| iaxflood               | iaxflood               |         |
| inviteflood            | inviteflood            |         |
| legion                 | legion                 |         |
| lynis                  | lynis                  |         |
| nikto                  | nikto                  |         |
| nmap                   | nmap                   |         |
| ohrwurm                | ohrwurm                |         |
| peass                  | peass                  |         |
| protos-sip             | protos-sip             |         |
| rtpbreak               | rtpbreak               |         |
| rtpflood               | rtpflood               |         |
| rtpinsertsound         | rtpinsertsound         |         |
| rtpmixsound            | rtpmixsound            |         |
| sctpscan               | sctpscan               |         |
| sfuzz                  | sfuzz                  |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| siege              | siege              |         |
| siparmyknife       | siparmyknife       |         |
| sipp               | sipp               |         |
| sipsak             | sipsak             |         |
| sipvicious         | sipvicious         |         |
| slowhttptest       | slowhttptest       |         |
| spike              | spike              |         |
| t50                | t50                |         |
| thc-ssl-dos        | thc-ssl-dos        |         |
| unix-privesc-check | unix-privesc-check |         |
| voiphopper         | voiphopper         |         |
| yersinia           | yersinia           |         |

## Web

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| apache-users         | apache-users         |         |
| apache2              | apache2              |         |
| beef-xss             | beef-xss             |         |
| burpsuite            | burpsuite            |         |
| cadaver              | cadaver              |         |
| commix               | commix               |         |
| cutycapt             | cutycapt             |         |
| davtest              | davtest              |         |
| default-mysql-server | default-mysql-server |         |
| dirb                 | dirb                 |         |
| dirbuster            | dirbuster            |         |
| dotdotpwn            | dotdotpwn            |         |
| eyewitness           | eyewitness           |         |
| ferret-sidejack      | ferret-sidejack      |         |
| ftester              | ftester              |         |
| hakrawler            | hakrawler            |         |
| hamster-sidejack     | hamster-sidejack     |         |
| heartleech           | heartleech           |         |
| httpprint            | httpprint            |         |

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| httrack         | httrack         |         |
| hydra           | hydra           |         |
| hydra-gtk       | hydra-gtk       |         |
| jboss-autopwn   | jboss-autopwn   |         |
| joomscan        | joomscan        |         |
| jsql-injection  | jsql-injection  |         |
| laudanum        | laudanum        |         |
| lbd             | lbd             |         |
| maltego         | maltego         |         |
| medusa          | medusa          |         |
| mitmproxy       | mitmproxy       |         |
| ncrack          | ncrack          |         |
| nikto           | nikto           |         |
| nishang         | nishang         |         |
| nmap            | nmap            |         |
| osscanner       | osscanner       |         |
| owasp-mantra-ff | owasp-mantra-ff |         |
| padbuster       | padbuster       |         |
| paros           | paros           |         |
| patator         | patator         |         |
| php             | php             |         |
| php-mysql       | php-mysql       |         |
| proxychains4    | proxychains4    |         |
| proxytunnel     | proxytunnel     |         |
| qsslcaudit      | qsslcaudit      |         |
| redsocks        | redsocks        |         |
| sidguesser      | sidguesser      |         |
| siege           | siege           |         |
| skipfish        | skipfish        |         |
| slowhttptest    | slowhttptest    |         |
| sqldict         | sqldict         |         |
| sqlitebrowser   | sqlitebrowser   |         |
| sqlmap          | sqlmap          |         |
| sqlninja        | sqlninja        |         |

| Tool        | Command(s)  | Purpose |
|-------------|-------------|---------|
| sqlsus      | sqlsus      |         |
| ssldump     | ssldump     |         |
| sslh        | sslh        |         |
| sslscan     | sslscan     |         |
| sslsniff    | sslsniff    |         |
| sslsplit    | sslsplit    |         |
| sslyze      | sslyze      |         |
| stunnel4    | stunnel4    |         |
| thc-ssl-dos | thc-ssl-dos |         |
| tlssled     | tlssled     |         |
| tnscmd10g   | tnscmd10g   |         |
| uniscan     | uniscan     |         |
| wafw00f     | wafw00f     |         |
| wapiti      | wapiti      |         |
| watobo      | watobo      |         |
| webacoo     | webacoo     |         |
| webscarab   | webscarab   |         |
| webshells   | webshells   |         |
| weevely     | weevely     |         |
| wfuzz       | wfuzz       |         |
| whatweb     | whatweb     |         |
| wireshark   | wireshark   |         |
| wpscan      | wpscan      |         |
| xsser       | xsser       |         |
| zaproxy     | zaproxy     |         |

## FLARE-VM

### All Packages (choco)

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| 010editor          | 010editor          |         |
| 7zip               | 7zip               |         |
| advanced-installer | advanced-installer |         |
| angr               | angr               |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| apimonitor           | apimonitor           |         |
| apktool              | apktool              |         |
| asar                 | asar                 |         |
| autoit-ripper        | autoit-ripper        |         |
| binaryninja          | binaryninja          |         |
| bindiff              | bindiff              |         |
| blobrunner           | blobrunner           |         |
| blobrunner64         | blobrunner64         |         |
| bytecodeviewer       | bytecodeviewer       |         |
| capa                 | capa                 |         |
| capa-explorer-web    | capa-explorer-web    |         |
| chrome.extensions    | chrome.extensions    |         |
| cmdr                 | cmdr                 |         |
| codetrack            | codetrack            |         |
| cryptotester         | cryptotester         |         |
| cutter               | cutter               |         |
| cyberchef            | cyberchef            |         |
| cygwin               | cygwin               |         |
| de4dot-cex           | de4dot-cex           |         |
| dependencywalker     | dependencywalker     |         |
| dex2jar              | dex2jar              |         |
| didier-stevens-beta  | didier-stevens-beta  |         |
| didier-stevens-suite | didier-stevens-suite |         |
| die                  | die                  |         |
| dll-to-exe           | dll-to-exe           |         |
| dnlib                | dnlib                |         |
| dnspyex              | dnspyex              |         |
| dotdumper            | dotdumper            |         |
| dotnet3.5            | dotnet3.5            |         |
| exeinfope            | exeinfope            |         |
| explorersuite        | explorersuite        |         |
| extreme_dumper       | extreme_dumper       |         |
| ezviewer             | ezviewer             |         |
| fakenet-ng           | fakenet-ng           |         |

| Tool                     | Command(s)               | Purpose |
|--------------------------|--------------------------|---------|
| fiddler                  | fiddler                  |         |
| file                     | file                     |         |
| floss                    | floss                    |         |
| garbageman               | garbageman               |         |
| ghidra                   | ghidra                   |         |
| goresym                  | goresym                  |         |
| gostringungarbler        | gostringungarbler        |         |
| hashmyfiles              | hashmyfiles              |         |
| hollowshunter            | hollowshunter            |         |
| hxd                      | hxd                      |         |
| ida.plugin.capa          | ida.plugin.capa          |         |
| ida.plugin.comida        | ida.plugin.comida        |         |
| ida.plugin.delphihelper  | ida.plugin.delphihelper  |         |
| ida.plugin.dereferencing | ida.plugin.dereferencing |         |
| ida.plugin.diaphora      | ida.plugin.diaphora      |         |
| ida.plugin.flare         | ida.plugin.flare         |         |
| ida.plugin.flare-emu     | ida.plugin.flare-emu     |         |
| ida.plugin.hashdb        | ida.plugin.hashdb        |         |
| ida.plugin.hrtng         | ida.plugin.hrtng         |         |
| ida.plugin.ifl           | ida.plugin.ifl           |         |
| ida.plugin.xray          | ida.plugin.xray          |         |
| ida.plugin.xrefer        | ida.plugin.xrefer        |         |
| idafree                  | idafree                  |         |
| idr                      | idr                      |         |
| ifpstools                | ifpstools                |         |
| ilspy                    | ilspy                    |         |
| innoextract              | innoextract              |         |
| innounp                  | innounp                  |         |
| internet_detector        | internet_detector        |         |
| ipython                  | ipython                  |         |
| isd                      | isd                      |         |
| js-beautify              | js-beautify              |         |
| js-deobfuscator          | js-deobfuscator          |         |
| keystone                 | keystone                 |         |

| Tool                       | Command(s)                              | Purpose |
|----------------------------|-----------------------------------------|---------|
| libraries.python3          | <code>libraries.python3</code>          |         |
| magika                     | <code>magika</code>                     |         |
| malware-jail               | <code>malware-jail</code>               |         |
| map                        | <code>map</code>                        |         |
| microsoft-office           | <code>microsoft-office</code>           |         |
| nasm                       | <code>nasm</code>                       |         |
| net-reactor-slayer         | <code>net-reactor-slayer</code>         |         |
| nmap                       | <code>nmap</code>                       |         |
| notepadplusplus            | <code>notepadplusplus</code>            |         |
| notepadpp.plugin.compare   | <code>notepadpp.plugin.compare</code>   |         |
| notepadpp.plugin.jstool    | <code>notepadpp.plugin.jstool</code>    |         |
| notepadpp.plugin.xmltools  | <code>notepadpp.plugin.xmltools</code>  |         |
| obfuscator-io-deobfuscator | <code>obfuscator-io-deobfuscator</code> |         |
| offvis                     | <code>offvis</code>                     |         |
| onenoteanalyzer            | <code>onenoteanalyzer</code>            |         |
| pdbresym                   | <code>pdbresym</code>                   |         |
| pdfstreamdumper            | <code>pdfstreamdumper</code>            |         |
| pe_unmapper                | <code>pe_unmapper</code>                |         |
| pebear                     | <code>pebear</code>                     |         |
| peid                       | <code>peid</code>                       |         |
| pesieve                    | <code>pesieve</code>                    |         |
| pestudio                   | <code>pestudio</code>                   |         |
| pkg-unpacker               | <code>pkg-unpacker</code>               |         |
| pma-labs                   | <code>pma-labs</code>                   |         |
| procdot                    | <code>procdot</code>                    |         |
| processdump                | <code>processdump</code>                |         |
| psnotify                   | <code>psnotify</code>                   |         |
| pycdas                     | <code>pycdas</code>                     |         |
| pycdc                      | <code>pycdc</code>                      |         |
| pylingual                  | <code>pylingual</code>                  |         |
| rat-king-parser            | <code>rat-king-parser</code>            |         |
| recaf                      | <code>recaf</code>                      |         |
| reg_export                 | <code>reg_export</code>                 |         |
| regcool                    | <code>regcool</code>                    |         |



| Tool                     | Command(s)               | Purpose |
|--------------------------|--------------------------|---------|
| regshot                  | regshot                  |         |
| resourcehacker           | resourcehacker           |         |
| rundotnetdll             | rundotnetdll             |         |
| scdbg                    | scdbg                    |         |
| sclauncher               | sclauncher               |         |
| sclauncher64             | sclauncher64             |         |
| sfextract                | sfextract                |         |
| shellcode_launcher       | shellcode_launcher       |         |
| sysinternals             | sysinternals             |         |
| systeminformer           | systeminformer           |         |
| ttd                      | ttd                      |         |
| uncompyle6               | uncompyle6               |         |
| uniextract2              | uniextract2              |         |
| unpyc3                   | unpyc3                   |         |
| upx                      | upx                      |         |
| vb-decompiler-lite       | vb-decompiler-lite       |         |
| vbdec                    | vbdec                    |         |
| vcbuildtools             | vcbuildtools             |         |
| vcredis-all              | vcredis-all              |         |
| vscode                   | vscode                   |         |
| vscode.extension.jupyter | vscode.extension.jupyter |         |
| vscode.extension.python  | vscode.extension.python  |         |
| windbg                   | windbg                   |         |
| windows-terminal         | windows-terminal         |         |
| wireshark                | wireshark                |         |
| x64dbg                   | x64dbg                   |         |
| x64dbg.plugin.dbgchild   | x64dbg.plugin.dbgchild   |         |
| x64dbg.plugin.ollydumpex | x64dbg.plugin.ollydumpex |         |
| x64dbg.plugin.scyllahide | x64dbg.plugin.scyllahide |         |
| x64dbg.plugin.x64dbgpy   | x64dbg.plugin.x64dbgpy   |         |
| yara                     | yara                     |         |

## SIFT Workstation

### Forensic Packages

| Tool                       | Command(s)                 | Purpose |
|----------------------------|----------------------------|---------|
| absent                     | absent                     |         |
| aeskeyfind                 | aeskeyfind                 |         |
| afflib-tools               | afflib-tools               |         |
| aircrack-ng                | aircrack-ng                |         |
| android-sdk-platform-tools | android-sdk-platform-tools |         |
| arp-scan                   | arp-scan                   |         |
| autopsy                    | autopsy                    |         |
| avfs                       | avfs                       |         |
| aws-cli                    | aws-cli                    |         |
| bless                      | bless                      |         |
| blt                        | blt                        |         |
| bulk-extractor             | bulk-extractor             |         |
| cabextract                 | cabextract                 |         |
| ccrypt                     | ccrypt                     |         |
| chromium-browser           | chromium-browser           |         |
| clamav                     | clamav                     |         |
| claude-code                | claude-code                |         |
| cmospwd                    | cmospwd                    |         |
| cryptcat                   | cryptcat                   |         |
| cryptsetup                 | cryptsetup                 |         |
| dc3dd                      | dc3dd                      |         |
| dcfldd                     | dcfldd                     |         |
| default-jre                | default-jre                |         |
| disktype                   | disktype                   |         |
| dislocker                  | dislocker                  |         |
| docker                     | docker                     |         |
| dos2unix                   | dos2unix                   |         |
| dotnet                     | dotnet                     |         |
| driftnet                   | driftnet                   |         |
| dsniff                     | dsniff                     |         |
| e2fsprogs                  | e2fsprogs                  |         |
| ent                        | ent                        |         |
| epic5                      | epic5                      |         |
| etherape                   | etherape                   |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| ettercap-graphical | ettercap-graphical |         |
| ewf-tools          | ewf-tools          |         |
| exfat-extras       | exfat-extras       |         |
| exfat-fuse         | exfat-fuse         |         |
| exif               | exif               |         |
| extundelete        | extundelete        |         |
| fdupes             | fdupes             |         |
| feh                | feh                |         |
| file               | file               |         |
| flex               | flex               |         |
| foremost           | foremost           |         |
| gawk               | gawk               |         |
| gdb                | gdb                |         |
| gddrescue          | gddrescue          |         |
| ghex               | ghex               |         |
| graphviz           | graphviz           |         |
| grepcidr           | grepcidr           |         |
| gthumb             | gthumb             |         |
| gzrt               | gzrt               |         |
| hashdeep           | hashdeep           |         |
| hexedit            | hexedit            |         |
| hydra              | hydra              |         |
| hydra-gtk          | hydra-gtk          |         |
| init               | init               |         |
| ipython3           | ipython3           |         |
| jq                 | jq                 |         |
| kdiff3             | kdiff3             |         |
| kpartx             | kpartx             |         |
| lft                | lft                |         |
| lvm2               | lvm2               |         |
| magnus             | magnus             |         |
| mdadm              | mdadm              |         |
| mtd-utils          | mtd-utils          |         |
| nbd-client         | nbd-client         |         |

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| nbtscan         | nbtscan         |         |
| netcat          | netcat          |         |
| netpbm          | netpbm          |         |
| netsed          | netsed          |         |
| netwox          | netwox          |         |
| nfdump          | nfdump          |         |
| ngrep           | ngrep           |         |
| nikto           | nikto           |         |
| ntfs-3g         | ntfs-3g         |         |
| okular          | okular          |         |
| onboard         | onboard         |         |
| open-iscsi      | open-iscsi      |         |
| openjdk         | openjdk         |         |
| ophcrack        | ophcrack        |         |
| ophcrack-cli    | ophcrack-cli    |         |
| orca            | orca            |         |
| outguess        | outguess        |         |
| p0f             | p0f             |         |
| p7zip-full      | p7zip-full      |         |
| patch           | patch           |         |
| pdftk-java      | pdftk-java      |         |
| perl            | perl            |         |
| pev             | pev             |         |
| pff-tools       | pff-tools       |         |
| phonon          | phonon          |         |
| pkg-config      | pkg-config      |         |
| plaso-tools     | plaso-tools     |         |
| powershell      | powershell      |         |
| pst-utils       | pst-utils       |         |
| pv              | pv              |         |
| python-flowgrep | python-flowgrep |         |
| python3-debian  | python3-debian  |         |
| python3-dfvfs   | python3-dfvfs   |         |
| python3-fuse    | python3-fuse    |         |

| Tool                    | Command(s)              | Purpose |
|-------------------------|-------------------------|---------|
| python3-keyrings-alt    | python3-keyrings-alt    |         |
| python3-m2crypto        | python3-m2crypto        |         |
| python3-magic           | python3-magic           |         |
| python3-pefile          | python3-pefile          |         |
| python3-plaso           | python3-plaso           |         |
| python3-pypff           | python3-pypff           |         |
| python3-pyqt5           | python3-pyqt5           |         |
| python3-pytsk3          | python3-pytsk3          |         |
| python3-redis           | python3-redis           |         |
| python3-setuptools-rust | python3-setuptools-rust |         |
| python3-tk              | python3-tk              |         |
| python3-tsk             | python3-tsk             |         |
| python3-virtualenv      | python3-virtualenv      |         |
| python3-wxgtk4          | python3-wxgtk4          |         |
| python3-xlsxwriter      | python3-xlsxwriter      |         |
| python3-yara            | python3-yara            |         |
| qemu                    | qemu                    |         |
| qemu-utils              | qemu-utils              |         |
| radare2                 | radare2                 |         |
| rar                     | rar                     |         |
| rsakeyfind              | rsakeyfind              |         |
| safecopy                | safecopy                |         |
| samdump2                | samdump2                |         |
| scalpel                 | scalpel                 |         |
| silversearcher-ag       | silversearcher-ag       |         |
| sleuthkit               | sleuthkit               |         |
| socat                   | socat                   |         |
| squashfs-tools          | squashfs-tools          |         |
| ssdeep                  | ssdeep                  |         |
| ssldump                 | ssldump                 |         |
| sslsniff                | sslsniff                |         |
| stunnel4                | stunnel4                |         |
| swig                    | swig                    |         |
| tcl                     | tcl                     |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| tcpflow              | tcpflow              |         |
| tcpick               | tcpick               |         |
| tcpreplay            | tcpreplay            |         |
| tcpslice             | tcpslice             |         |
| tcpstat              | tcpstat              |         |
| tcptrace             | tcptrace             |         |
| tcptrack             | tcptrack             |         |
| tcpxtract            | tcpxtract            |         |
| testdisk             | testdisk             |         |
| tofrodo              | tofrodo              |         |
| transmission         | transmission         |         |
| ugrep                | ugrep                |         |
| unity-control-center | unity-control-center |         |
| unrar                | unrar                |         |
| upx-ucl              | upx-ucl              |         |
| vbindiff             | vbindiff             |         |
| virtuoso-minimal     | virtuoso-minimal     |         |
| vmfs-tools           | vmfs-tools           |         |
| winbind              | winbind              |         |
| wine                 | wine                 |         |
| wireshark            | wireshark            |         |
| xdot                 | xdot                 |         |
| xfsprogs             | xfsprogs             |         |
| xmount               | xmount               |         |
| zenity               | zenity               |         |
| zlib1g-dev           | zlib1g-dev           |         |

## Security Onion

### DB/Search

| Tool          | Command(s) | Purpose                                             |
|---------------|------------|-----------------------------------------------------|
| Elasticsearch | —          | NoSQL index for logs/metadata/alerts                |
| ILM           | —          | Index Lifecycle Management (aging/rolling/deletion) |
| Kibana        | —          | Search analytics & dashboards                       |

## Detection

| Tool              | Command(s) | Purpose                                           |
|-------------------|------------|---------------------------------------------------|
| ATT&CK Navigator  | —          | Visual defensive-coverage matrix                  |
| ElastAlert 2      | —          | Continuous alert-indicator scanner                |
| Playbook          | —          | Custom detections mapped to MITRE ATT&CK          |
| Sigma / Sigma-CLI | —          | Translate YAML detection rules to Elastic queries |
| so-idstools       | —          | Manage/update Suricata ET rules                   |

## Diagnostics

| Tool     | Command(s) | Purpose                                |
|----------|------------|----------------------------------------|
| Grafana  | —          | Infra/node-health dashboards           |
| InfluxDB | —          | Time-series store for hardware metrics |
| Telegraf | —          | Hardware health/CPU/mem/disk agent     |

## File Analysis

| Tool        | Command(s) | Purpose                                       |
|-------------|------------|-----------------------------------------------|
| DomainStats | —          | Domain-age checks to flag malicious sites     |
| FreqServer  | —          | Entropy analysis to detect DGAs               |
| Strelka     | —          | Real-time file extraction & YARA/file carving |

## Host/EDR

| Tool          | Command(s) | Purpose                                           |
|---------------|------------|---------------------------------------------------|
| Elastic Agent | —          | Host log/metric/audit collector                   |
| Elastic Fleet | —          | Central agent config & deployment                 |
| OpenCanary    | —          | Network honeypot / decoy services                 |
| Osquery       | —          | Live endpoint queries via SQL                     |
| Wazuh         | —          | HIDS - File Integrity Monitoring & rootkit checks |

## IR/AI

| Tool                         | Command(s) | Purpose                                     |
|------------------------------|------------|---------------------------------------------|
| CyberChef                    | —          | Visual data decode/parse/analysis           |
| Onion AI Assistant           | —          | AI triage assistance for analysts           |
| Security Onion Console (SOC) | —          | Web UI for threat hunting & grid management |
| TheHive                      | —          | Collaborative IR & case management          |

## Identity/Infra

| Tool             | Command(s) | Purpose                                        |
|------------------|------------|------------------------------------------------|
| ManagerHype      | —          | Hypervisor integration for VM mapping          |
| Nginx (so-nginx) | —          | Reverse proxy for internal web traffic         |
| Ory Kratos & Dex | —          | IAM - SSO/auth/sessions                        |
| so-apt-cacher-ng | —          | Local package caching proxy                    |
| so-firewall      | —          | iptables/nftables wrapper for inter-node comms |

## Log Routing

| Tool                | Command(s) | Purpose                                   |
|---------------------|------------|-------------------------------------------|
| Filebeat            | —          | Lightweight log shipper                   |
| Logstash            | —          | Data pipeline - ECS normalize & enrich    |
| Redis               | —          | Log ingestion queue/broker                |
| Rsyslog / Syslog-ng | —          | Ingest traditional syslog from appliances |

## Network Visibility

| Tool         | Command(s) | Purpose                                              |
|--------------|------------|------------------------------------------------------|
| Sensoroni    | —          | Backend API agent retrieving/extracting PCAPs        |
| Stenographer | —          | High-speed raw full packet capture daemon            |
| Suricata     | —          | Signature-based IDS/IPS & packet capture             |
| Zeek         | —          | Network traffic analyzer & protocol metadata logging |

## Sources

- `remnux_docs` — <https://raw.githubusercontent.com/REMnux/docs/master/discover-the-tools>
- `kali_control` — <https://gitlab.com/kalilinux/packages/kali-meta/-/raw/kali/master/debian/control>
- `flare_config` — <https://raw.githubusercontent.com/mandiant/flare-vm/main/config.xml>



# Tool Reference

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One page per tool. Every option below was read off the real binary and is checked by the linter.

# fls

**Kit:** SIFT Workstation · REMnux · Kali ( `sleuthkit` ) **Category:** Filesystem analysis **Version:** The Sleuth Kit 4.11.1  
**Docs:** <https://wiki.sleuthkit.org/index.php?title=Fls> **Verified:** 2026-07-29 from `cyberlab-aio:v1` — raw output in `capture/fls.help.txt`

**Worked example of the format.** Every option below was read off the real binary, not written from memory.  
See [docs/FORMAT.md](#).

## Purpose

List file and directory names in a disk image, including deleted entries.

## Synopsis

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]  
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

If `[inode]` is not given, the root directory is used.

**Note:** the usage line above advertises only `[-adDFlhpruvV]`, but the tool also accepts `-P`, `-B`, `-S` and `-k`. They are documented below.

## Common invocations

```
# List the root directory of an image
fls {{path/to/image.dd}}

# List a partition that starts at a sector offset (the usual case for a full disk)
fls -o {{2048}} {{path/to/image.dd}}

# Show deleted entries only – the fast answer to "what was removed?"
fls -d -o {{2048}} {{path/to/image.dd}}

# Recurse the whole filesystem showing full paths, deleted files included
fls -rp -o {{2048}} {{path/to/image.dd}}

# Recurse showing ONLY deleted files with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}

# Long listing (sizes, MAC times) like ls -l, in a named time zone
fls -l -z {{EST5EDT}} -o {{2048}} {{path/to/image.dd}}

# Emit a body file for timelines, then build the timeline with mactime
fls -r -m {{/}} -o {{2048}} {{path/to/image.dd}} > {{path/to/body.txt}}

# List the contents of one directory by inode number
fls -o {{2048}} {{path/to/image.dd}} {{inode}}
```

## Options — complete

| Flag            | Argument | What it does                                       | When you would use it                                              |
|-----------------|----------|----------------------------------------------------|--------------------------------------------------------------------|
| <code>-a</code> | —        | Display <code>.</code> and <code>..</code> entries | Rarely; they are hidden by default as noise                        |
| <code>-d</code> | —        | Display deleted entries only                       | Recovery hunts — isolates what was removed                         |
| <code>-D</code> | —        | Display only directories                           | Mapping structure without file noise                               |
| <code>-F</code> | —        | Display only files                                 | Excluding directories from a listing                               |
| <code>-l</code> | —        | Display long version (like <code>ls -l</code> )    | When you need sizes, UID/GID and MAC times                         |
| <code>-p</code> | —        | Display full path for each file                    | Almost always with <code>-r</code> ; otherwise output is ambiguous |
| <code>-r</code> | —        | Recurse on directory entries                       | Walking a whole filesystem rather than one directory               |
| <code>-u</code> | —        | Display undeleted entries only                     | The inverse of <code>-d</code> ; filtering live files              |

| Flag            | Argument                       | What it does                                                                      | When you would use it                                                             |
|-----------------|--------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <code>-m</code> | <code>dir/</code>              | Output in mactime input format, with <code>dir/</code> as the mount point         | Building a super-timeline; feeds <code>mactime</code>                             |
| <code>-h</code> | —                              | Include MD5 checksum hash in mactime output                                       | Timeline work where you also need file hashes                                     |
| <code>-f</code> | <code>fstype</code>            | Filesystem type ( <code>-f list</code> for supported types)                       | When auto-detection fails or is wrong                                             |
| <code>-i</code> | <code>imgtype</code>           | Image format ( <code>-i list</code> for supported types)                          | Non-raw images: E01, AFF, etc.                                                    |
| <code>-b</code> | <code>dev_sector_size</code>   | Device sector size in bytes                                                       | 4Kn drives, where the 512-byte assumption breaks                                  |
| <code>-o</code> | <code>imgoffset</code>         | Offset into the image, in sectors                                                 | <b>The most common flag</b> — targeting a partition in a full-disk image          |
| <code>-P</code> | <code>pooltype</code>          | Pool container type ( <code>-P list</code> for supported types)                   | APFS / logical volume containers                                                  |
| <code>-B</code> | <code>pool_volume_block</code> | Starting block, pool volumes only                                                 | Selecting a volume inside a pool                                                  |
| <code>-S</code> | <code>snap_id</code>           | Snapshot ID (APFS only)                                                           | Examining a specific APFS snapshot                                                |
| <code>-k</code> | <code>password</code>          | Decryption password for encrypted volumes                                         | BitLocker/FileVault-style encrypted volumes                                       |
| <code>-z</code> | <code>ZONE</code>              | Time zone of the original machine (e.g. <code>EST5EDT</code> , <code>GMT</code> ) | Only meaningful with <code>-l</code> ; wrong TZ silently skews every timestamp    |
| <code>-s</code> | <code>seconds</code>           | Time skew of the original machine, in seconds                                     | Only with <code>-l</code> / <code>-m</code> ; correcting a known-bad system clock |
| <code>-v</code> | —                              | Verbose output to stderr                                                          | Debugging why an image will not parse                                             |
| <code>-V</code> | —                              | Print version                                                                     | Recording tool provenance in your notes                                           |

## Gotchas

- `-o` **is in sectors, not bytes**. Get it from `mm1s` first. A wrong offset usually yields "Cannot determine file system type", not an obvious error.
- `-z` **only affects** `-l` **output**. Setting it without `-l` changes nothing, which makes a wrong-timezone timeline easy to produce and hard to notice.
- **Deleted entries may have unrecoverable content**. `fls -d` listing a name does not mean `icat` will return its data — the blocks may be reallocated.
- `-r` **on a large image is slow and noisy**. Pipe to a file, then `grep` it, rather than scrolling.

## See also

- `mmls` — partition table; where you get the `-o` value
- `icat` — extract the content of an inode `fls` found
- `istat` — detailed metadata for one inode
- `mactime` — consumes the `-m` body file to build the timeline
- `tsk_recover` — bulk export of allocated/deleted files